

Problem I

Marks

Time limit: 1 second

Problem Description

There is a circular track with b equally spaced positions, numbered $0, 1, \dots, b-1$ clockwise. Adjacent positions are 1 unit apart, so the whole circle measures b units.

A marker starts at position 0 and, at every step, advances by a fixed a positions clockwise (wrapping back to 0 after position $b-1$). It successively stops at

$$p_k = (a \cdot k) \bmod b, \quad k = 0, 1, \dots, n-1,$$

leaving a mark at each stop, for a total of n marks. It is guaranteed that $\gcd(a, b) = 1$ ¹ and $n \leq b$, so these n marks are pairwise distinct.

These n marks divide the circle into n arcs. The length of an arc is the number of units it spans (every arc length is a positive integer, and the lengths sum to b).

How many **distinct** arc-length values are there? For each value, what is it, and how many times does it occur?

Input Format

Each test contains multiple test cases. The first line contains the number of test cases t . The description of the test cases follows.

The only line of each test case contains three integers a , b , and n , representing the parameters of this test case.

Output Format

For each test case, first output the number of distinct arc-length values, d .

Followed by d lines, output 2 integers s and c for each of which, representing that exactly c arcs have length s .

For each test case, output these lines **in increasing order of s** .

¹ $\gcd(a, b) = c$ means that c is the largest positive integer such that c divides both a and b .

Technical Specification

- $1 \leq t \leq 30$
- $1 \leq a < b \leq 10^{18}$
- $\gcd(a, b) = 1$
- $2 \leq n \leq b$

Sample Input 1

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1
7 12 7
```

Sample Output 1

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2
1 2
2 5
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Explanation for Sample 1: The marks are $\{0, 7, 2, 9, 4, 11, 6\}$, which, when sorted, are 0, 2, 4, 6, 7, 9, 11. The consecutive arc lengths are 2, 2, 2, 1, 2, 2, 1, so length 1 occurs 2 times and length 2 occurs 5 times.